

BOLOGNA, Italy

WMO: 161400

Lat: 44.5275N

Lon: 11.2886E

Elev: 123

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.9	(c) 28.0	(d) 14.1	(e) 11.3	(f) 32.8	(g) 17.7	(h) 13.5	(i) 34.0	(j) 17.7	(k) 44.2	(l) 14.2	(m) 42.5	(n) 5.1	(o) 210	(p) 0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	94.8	72.6	91.7	72.0	89.5	71.3	76.0	88.4	74.3	86.8	72.8	85.2	6.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	118.5	82.8	70.1	111.4	80.9	68.4	104.9	79.6	39.5	88.6	37.8	87.2	36.5	85.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.9	(b) 13.6	(c) 11.8	(e) 20.4	(f) 98.4	(g) 4.0	(h) 2.8	(i) 17.5	(j) 100.4	(k) 15.1	(l) 102.0	(m) 12.8	(n) 103.6	(o) 9.9	(p) 105.7		
(a) 15.9	(b) 13.6	(c) 11.8	(e) 18.9	(f) 77.9	(g) 3.8	(h) 1.9	(i) 16.2	(j) 79.2	(k) 13.9	(l) 80.3	(m) 11.8	(n) 81.4	(o) 9.0	(p) 82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	38.7	41.9	49.8	57.0	65.3	73.4	78.0	77.4	68.6	59.5	49.0	39.9
	DBStd	14.94	4.96	5.99	6.42	5.63	5.37	5.79	4.54	5.14	5.51	6.11	6.40	5.48
	HDD50	1085	352	229	81	10	0	0	0	0	0	6	92	315
	HDD65	3721	816	646	471	245	62	7	0	0	25	190	481	778
	CDD50	4120	1	4	76	220	476	703	869	850	558	300	61	2
	CDD65	1279	0	0	0	5	72	261	404	385	133	19	0	0
	CDH74	12413	0	0	5	48	563	2478	4301	3978	974	66	0	0
	CDH80	5061	0	0	0	3	114	937	1945	1797	260	5	0	0
	WSAvg	5.2	4.5	4.8	5.5	5.9	5.9	5.9	5.9	5.6	5.3	4.6	4.6	4.3
	PrecAvg	28.5	1.7	1.8	2.4	2.8	2.5	2.2	1.9	2.2	2.4	2.8	3.1	2.5
(15) Precipitation	PrecMax	42.0	5.8	5.7	6.8	6.5	5.4	4.7	5.5	4.8	7.6	11.1	6.5	5.2
	PrecMin	16.5	0.0	0.0	0.0	0.4	0.0	0.5	0.1	0.1	0.0	0.1	0.1	0.4
	PrecStd	6.3	1.4	1.4	1.3	1.8	1.5	1.2	1.5	1.5	1.9	2.4	1.7	1.3
	PrecStd	6.3	1.4	1.4	1.3	1.8	1.5	1.2	1.5	1.5	1.9	2.4	1.7	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	62.4	73.1	79.1	87.5	94.9	98.0	98.9	89.8	79.8	68.3	57.4
		MCWB	48.4	49.6	56.8	62.8	68.4	71.9	71.4	72.0	70.5	65.8	59.3	50.5
	2%	DB	52.1	57.5	67.9	74.8	82.8	91.5	94.7	95.1	86.2	75.4	64.0	52.8
		MCWB	45.2	48.2	54.5	60.0	66.2	71.9	72.7	72.8	69.4	64.4	57.5	48.0
	5%	DB	48.8	55.1	64.7	71.5	80.2	88.2	91.8	91.7	82.8	72.3	60.6	50.1
		MCWB	43.9	47.1	53.0	58.4	65.0	70.8	72.0	72.2	67.7	62.8	55.8	46.6
	10%	DB	46.3	51.8	61.4	68.1	77.0	85.7	89.5	89.3	79.8	69.4	57.6	48.0
		MCWB	42.7	45.3	51.4	56.7	63.5	69.5	71.3	71.4	65.9	61.2	54.0	45.1
	0.4%	WB	50.2	52.1	58.0	63.9	70.7	75.8	78.1	78.7	73.3	68.6	61.3	52.9
		MCDB	54.3	59.3	68.3	76.2	82.3	88.4	91.9	91.8	83.9	74.5	64.5	56.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.0	49.8	55.8	61.3	68.1	73.6	75.8	76.4	71.4	66.2	58.8	49.4
		MCDB	49.9	55.5	65.1	71.6	79.7	86.9	88.6	88.4	81.4	72.1	62.7	50.9
	5%	WB	44.7	48.0	54.2	59.4	66.3	72.2	74.3	74.6	69.7	64.1	56.9	47.1
		MCDB	47.1	53.1	62.8	69.1	76.9	85.1	86.8	86.8	79.7	70.5	60.0	49.0
	10%	WB	42.9	46.2	52.4	57.7	64.6	70.6	72.7	72.9	67.8	62.3	54.8	45.0
		MCDB	45.4	50.1	59.9	66.2	74.6	82.9	85.4	85.0	77.2	68.4	57.8	47.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	14.6	17.7	17.6	18.4	19.3	20.2	20.3	18.2	14.4	11.2	10.9
		MCDBR	16.8	19.9	23.1	22.5	22.2	22.6	22.7	23.2	21.4	17.8	14.7	14.3
		MCWBR	11.8	12.7	13.1	11.9	10.9	9.9	9.7	9.5	9.3	9.3	9.6	10.3
	5% WB	MCDBR	13.8	17.5	20.3	19.9	19.8	20.7	20.4	20.5	18.6	15.5	12.3	11.4
		MCWBR	10.5	11.9	12.4	11.1	10.6	10.0	9.9	9.7	9.1	8.6	8.5	8.8
(40) Clear-Sky Solar Irradiance	taub		0.369	0.396	0.443	0.465	0.463	0.478	0.466	0.455	0.442	0.442	0.399	0.363
	taud		2.308	2.198	2.078	2.057	2.109	2.118	2.163	2.211	2.214	2.167	2.227	2.344
	Ebn at Noon		229	243	247	253	258	254	256	254	246	224	213	219
	Edh at Noon		27	35	45	50	49	49	46	43	39	36	29	24
(44) All-Sky Solar Radiation	RadAvg		438	708	1113	1459	1821	2076	2139	1831	1338	816	470	384
	RadStd		68	105	156	146	158	120	114	117	94	104	64	50

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) -1.75	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (25 neighbors)		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+121	N/A	N/A

Nomenclature: See separate page